FOREX TRADING

The Forex Trading Playbook

Strategy Development, Backtesting & Journaling Systems

Volume 4 — Strategy & Systems

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Edition 2026

Contents

Read in order on your first pass — each chapter builds on the last — then keep this as a reference.

FRONT MATTER

Introduction

Strategy Development, Backtesting & Journaling Systems

Volumes 1–3 gave you knowledge: how markets work, how the best traders think, how to read a chart. This volume answers the question that turns a *student* into a *trader*:

> **"How do I know if what I'm doing actually works — before I bet real money on it?"**

The answer is a **system**: a precisely defined strategy, tested against history, validated forward on demo, and tracked through a rigorous journal so that *data*, not hope, tells you whether you have an edge. This is the engineering volume — the unglamorous machinery that separates consistent professionals from gamblers with a good story.

> ⚠️ **The hard truth this volume exists to enforce.** Most traders never actually *know* if their strategy is profitable. They trade on vibes, remember their wins, forget their losses, and tweak randomly after every drawdown. This volume replaces all of that with a **repeatable, measurable process.** It is less exciting than chart patterns — and far more important to your survival. A mediocre strategy with rigorous testing and journaling beats a "brilliant" one you've never measured.

This is **Volume 4**. It builds directly on **Vol 1 — Foundations** (risk, expectancy), **Vol 2 — Secrets of the Masters** (process, the business of trading), and **Vol 3 — Technical Analysis** (the raw material for your setups).

The development loop (the whole volume in one diagram)

```
┌──────────────────────────────────────────────────────────────┐
│  1. HYPOTHESIZE   →  An idea for an edge ("X tends to lead to Y")
│  2. SPECIFY       →  Define it precisely enough to test
│  3. BACKTEST      →  Run it on history; measure expectancy
│  4. FORWARD-TEST  →  Validate on demo / out-of-sample data
│  5. TRADE LIVE    →  Small, with strict risk; JOURNAL everything
│  6. ANALYZE       →  Compute KPIs; find what works & what leaks
│  7. REFINE        →  Change ONE variable; re-validate
└──────────────────────────────── ↺  loops forever  ──────────┘
```

> 🔑 **Trading is a business of edges measured over large samples.** No single trade matters; the *process* of finding, validating, and refining a positive-expectancy edge is the entire job. This volume is that process.

CHAPTER 1

From Knowledge to a Trading System

Knowing *about* trading and *having a system* are completely different things. You can understand every candlestick pattern and still lose money, because knowledge without a defined, repeatable system is just opinion. This chapter defines what a **trading system** actually is and why it's the foundation of everything that follows.

1.1 What is a trading system?

A **trading system** is a complete, pre-defined set of rules that answers — *before* you ever click — every question a trade poses:

```
WHAT    do I trade?            (markets / instruments)
WHEN    do I look?            (timeframes, sessions, conditions)
WHY     would I enter?        (the edge / setup criteria)
WHERE   do I enter?           (precise entry trigger)
HOW MUCH do I risk?           (position sizing)
WHERE   do I exit a loss?     (stop loss / invalidation)
WHERE   do I exit a win?      (targets / trailing / scaling)
HOW     do I record & review? (journaling & analytics)
```

If any of these is decided *in the moment*, emotionally, you don't have a system — you have a habit of gambling. A system makes the decisions in advance, in a calm state, so that execution becomes mechanical and measurable.

> 🔑 **The defining test of a system:** could you hand your rules to another disciplined person and have them take the *same* trades you would? If not, your rules aren't defined enough yet.

1.2 Strategy vs system vs plan — clearing up the terms

These words get used loosely. Here's a clean hierarchy:

Term	What it is	Scope
Setup / Strategy	A specific edge with entry & exit rules (e.g. "trend pullback to demand zone")	One repeatable trade type
Trading System	One or more strategies + risk rules + execution rules, fully specified	How you trade
Trading Plan	The system + your goals, schedule, psychology rules, and routines	Your whole operation (Vol 1 template)

This volume is mostly about building and validating the **strategy** and **system** layers — and the journaling that proves they work.

1.3 Mechanical vs discretionary systems

There's a spectrum between two poles:

Fully mechanical (rule-based / algorithmic)

Every decision is a hard rule a computer could execute. *"Buy when the 50 EMA crosses above the 200 EMA and RSI > 50; stop at 1.5× ATR; target 3R."*
- ✅ **Backtestable precisely**, removes emotion, perfectly consistent, scalable to code.
- ❌ Rigid; can miss context; needs robust testing to avoid curve-fitting.

Discretionary (judgment-based)

The trader applies experience and reads context within a framework. *"I look for high-confluence pullbacks in strong trends and judge entry quality."*
- ✅ Adapts to context, reads nuance a rule can't.
- ❌ Hard to test, hard to stay consistent, vulnerable to emotion and bias.

The pragmatic middle: "rules-based discretionary"

Most successful retail traders live here: a **mechanical skeleton** (defined setups, hard risk rules, fixed sizing) with **limited discretion** on entry quality. The rules are tight enough to **measure and journal**, but allow judgment where it genuinely adds value.

> 🔑 **The more discretionary your system, the more disciplined your journaling must be** — because the journal is the only way to measure something you can't fully codify. Discretion is fine; *unmeasured* discretion is not.

1.4 The non-negotiable components of any system

Whether mechanical or discretionary, every viable system must specify all of these (Chapter 2 turns each into a fillable spec):

```
1. MARKET / INSTRUMENT  — what you trade and why it suits the strategy
2. TIMEFRAME(S)         — analysis & execution timeframes
3. CONDITION FILTER     — when the strategy is valid (trending? ranging? session?)
4. ENTRY RULES          — the exact, repeatable trigger
5. EXIT RULES (LOSS)    — stop placement / invalidation
6. EXIT RULES (PROFIT)  — target(s), trailing, scaling
7. RISK & SIZING        — % risk per trade, position-size formula
8. RECORD & REVIEW      — what gets journaled and how it's measured
```

A system missing any one of these has a hole that *will* leak money — usually the exit and risk components, which beginners neglect in favor of entries.

> ⚠️ **Beginners obsess over entries; professionals obsess over exits and sizing.** Entries get you into the game; **exits and position sizing determine whether you win it.** A great entry with no exit/sizing plan is worthless.

1.5 Why you need a *system* (not just good trades)

- **Measurability:** You can only improve what you can measure. A system produces consistent, comparable data; random trading produces noise.

- **Consistency:** An edge only plays out over *many* trades taken the *same way*. Changing your approach every week guarantees you never sample your edge long enough to see it.
- **Emotional protection:** Rules decided in advance shield you from in-the-moment fear and greed (Vol 2, Ch. 6).
- **Falsifiability:** A defined system can be *proven wrong* by data — which means it can also be proven *right*. Vague trading can never be validated, so you're forever guessing.

> 🔑 The goal of this volume isn't to find a "perfect" strategy (none exists). It's to build a **defined, measurable system** so that you can *know your numbers* — and a known, positive-expectancy edge executed with discipline is the entire basis of professional trading.

1.6 Chapter summary

```
• A trading SYSTEM pre-answers every trade question so execution is mechanical & measurable.
• Hierarchy: Strategy/Setup ⊂ System ⊂ Trading Plan.
• Systems range from fully MECHANICAL to DISCRETIONARY; most pros sit in
  "rules-based discretionary" — a mechanical skeleton with limited judgment.
• Every system MUST specify: market, timeframe, condition filter, entry, stop,
  target, risk/sizing, and record/review. No exceptions.
• Obsess over EXITS and SIZING, not just entries.
• You need a system because only a consistent, measurable process lets an edge
  prove itself over a large sample.
```

Next: how to take a raw trading idea and forge it into a precise, testable strategy specification.

CHAPTER 2

Designing a Strategy

A strategy starts as an **idea** and must become a **specification** precise enough to test, journal, and execute identically every time. This chapter is the workshop where that transformation happens.

2.1 Start with a hypothesis (an edge you can state in one sentence)

Every strategy is really a **hypothesis about market behavior** — a claim that *"when condition X occurs, price tends to do Y often enough to be profitable after costs."* Examples:

- *"In a strong daily uptrend, pullbacks to a fresh demand zone tend to resume the trend."*
- *"After London-session consolidation, a break of the range tends to run during the New York session."*
- *"Following a bullish RSI divergence at a major support level, price tends to reverse upward."*

> 🔑 If you can't state your edge in a single clear sentence, you don't understand it well enough to test it. **Vagueness is the enemy of validation.** "I buy when it looks good" is not a hypothesis.

Good hypotheses come from: the technical concepts in Vol 3, observed market behavior, the principles of the masters in Vol 2, or a logical cause (e.g. session liquidity, central-bank themes from Vol 1). Avoid hypotheses with **no logical reason** behind them — random correlations don't survive forward into the future.

2.2 The strategy specification — turn the idea into rules

Take your hypothesis and define **every** component until it's unambiguous. This is your **strategy spec sheet** (full template in Chapter 9):

```
STRATEGY NAME:   ____________________________________

1. HYPOTHESIS / EDGE (one sentence):
   _________________________________________________

2. MARKET(S):        Which pairs, and why they suit this edge
3. TIMEFRAMES:       Bias TF + setup TF + entry TF
4. CONDITION FILTER: When is this strategy VALID / INVALID?
                     (e.g. ADX>25 trend only; specific session; not around red news)

5. ENTRY RULES (exact & repeatable):
   • Trigger 1: ____________________________________
   • Trigger 2: ____________________________________
   • Everything that must be TRUE to enter

6. STOP-LOSS RULE:   Where & why (structure / ATR-based); the invalidation point
7. TAKE-PROFIT RULE: Target(s), trailing, scaling-out logic
8. MIN RISK:REWARD:  e.g. ≥ 1:2

9. RISK PER TRADE:   ____ % (≤1%); position-size formula
10. TRADE MANAGEMENT: break-even rule, partials, max hold time
11. WHAT INVALIDATES THE WHOLE STRATEGY: when do you STOP trading it?
```

The test of a good spec (from Ch. 1): a disciplined stranger could read it and take the same trades you would.

2.3 The building blocks, in detail

Market & instrument selection

- Match the pair to the edge. A breakout strategy wants volatile, trending pairs; a range strategy wants ones that respect levels. Spreads/costs matter (Vol 1) — a scalp edge dies on wide-spread exotics.
- **Start narrow:** 1–3 pairs. You can't meaningfully test or monitor twenty.

Timeframe selection

- Pick a **bias TF**, a **setup TF**, and an **entry TF** (~4–6× apart — Vol 3, Ch. 8). The combination must fit your available screen time and personality (swing vs day vs scalp — Vol 1).

Condition filters — *when NOT to trade*

This is where amateurs leave money on the table. A strategy that's brilliant in trends is a disaster in ranges. Define the **regime** it needs:
- Trend vs range (e.g. ADX, structure, MA slope).
- Session/time-of-day (liquidity).
- News blackout windows (avoid high-impact releases).

> 🔑 A huge portion of edge comes from **not** trading the strategy when conditions are wrong. The filter is as important as the entry.

Entry rules — the trigger

Define the *exact* event that puts you in: e.g. *"a bullish engulfing candle closing inside the demand zone, with the entry-TF making a CHoCH up."* Specify whether you enter on the close, on a retest, or via a limit order at a level.

Exit rules — stop & target (the part that actually pays)

- **Stop:** at the **invalidation point** — the price that proves the hypothesis wrong (beyond structure / a multiple of ATR), with a buffer (Vol 3).
- **Target:** logical level / measured move / Fib extension, or a multiple of R. Define trailing and scaling rules explicitly.

Risk & sizing

Fixed % risk (≤1%) and the position-size formula (Vol 1; sizing models deepened in Ch. 8). This is constant across the system, independent of the setup.

2.4 Keep it simple — complexity is fragility

A frequent beginner mistake is piling on conditions ("RSI < 30 AND MACD cross AND 3 MAs aligned AND a doji AND Fibonacci AND...") believing more rules = more accuracy.

The opposite is usually true:
- **More parameters = more curve-fitting** (Chapter 4). A rule with ten conditions can be tuned to fit the past

perfectly and fail completely in the future.
- **Simple, logical strategies are more robust** — they capture a *real* behavior rather than memorizing historical noise.
- Each added rule needs *more* data to validate and reduces the number of trades, weakening your statistics.

> 🔑 **Aim for the fewest rules that capture a genuine edge.** Robustness beats complexity. If a strategy needs five exotic filters to look good in backtest, it's probably fitting noise, not finding an edge.

2.5 Define how you'll measure success *before* testing

Decide your acceptance criteria *now*, so you can't move the goalposts later (a classic self-deception):

```
Before testing, write down the MINIMUMS you'll require to trade it live:
  • Positive EXPECTANCY over the sample (Ch. 3)
  • Profit factor ≥ ____ (e.g. 1.3+)
  • Max drawdown ≤ ____ % (tolerable for you)
  • Minimum sample size ≥ ____ trades (e.g. 100)
  • Acceptable on OUT-OF-SAMPLE data, not just in-sample (Ch. 4)
```

Pre-committing to these numbers is what keeps backtesting *honest* (Chapter 4 is all about the ways we fool ourselves).

2.6 Chapter summary

```
• Every strategy = a one-sentence HYPOTHESIS about market behavior, with a logical reason.
• Forge it into a precise SPEC SHEET covering market, timeframes, condition filter,
  entry, stop, target, R:R, risk/sizing, management, and invalidation.
• A big edge lies in the CONDITION FILTER — knowing when NOT to trade it.
• Obsess over exits & sizing as much as entries.
• Keep it SIMPLE: fewer, logical rules are more robust; complexity = curve-fitting.
• Pre-commit to your success criteria (expectancy, profit factor, drawdown, sample)
  BEFORE you test, so you can't fool yourself later.
```

You have a defined strategy. Before testing it, you need to understand the **math that decides whether any strategy is actually profitable** — that's Chapter 3.

CHAPTER 3

Edge, Expectancy & the Math

This chapter is the **mathematical heart** of the whole series. Everything — your strategy, your backtest, your journal — exists to answer one question: *does this have a positive edge?* Master expectancy and you'll understand profitability more deeply than most retail traders ever do. The math is simple arithmetic; the implications are profound.

3.1 What "edge" actually means

An **edge** is a *statistical* advantage: a system whose average outcome per trade is positive after costs. It does **not** mean winning often, predicting the future, or being right on the next trade. It means that if you take the trade many times, the math is in your favor.

> 🔑 A casino has an edge of a few percent on roulette. It loses *individual* spins constantly — and is wildly profitable, because it plays the edge over millions of bets. **You are the casino, not the gambler.** Your job is to find a positive edge and execute it enough times for the math to express itself.

3.2 R-multiples — the universal unit (recap from Vol 1)

Express every result in **R**, where **1R = the amount you risked** on that trade.
- Risk \$50, make \$150 → **+3R**. Risk \$50, lose it → **−1R**.
- R standardizes everything regardless of account size or position size, so you can compare and aggregate trades cleanly. **All the math below uses R.**

3.3 Expectancy — the single most important number ⭐

Expectancy is your average profit (or loss) **per trade**, in R. It is the definition of whether you have an edge.

```
Expectancy (R) = (Win% × Average Win in R) − (Loss% × Average Loss in R)
```

Worked example

A system wins 40% of the time. Winners average +2.5R; losers average −1R.

```
Expectancy = (0.40 × 2.5R) − (0.60 × 1.0R)
           = 1.00R − 0.60R
           = +0.40R per trade
```

+0.40R expectancy means that, on average, every trade earns you 0.4× your risk. Over 100 trades risking 1% each, that's roughly **+40R** of edge expressed (before compounding and variance). A **positive expectancy is the mathematical proof of an edge.** Negative expectancy means no position sizing or psychology can save the system — it must be fixed or abandoned.

3.4 The win-rate trap: why win% alone is meaningless

Beginners chase high win rates. But win rate is **half** of the equation — and the less important half. Win rate and reward:risk **trade off against each other**:

Win rate	Avg win	Avg loss	Expectancy (R)	Verdict
90%	+0.5R	−5R	(0.9×0.5) − (0.1×5) = **−0.05R**	**Loses** despite 90% wins
50%	+2R	−1R	(0.5×2) − (0.5×1) = **+0.50R**	Strong edge
35%	+3R	−1R	(0.35×3) − (0.65×1) = **+0.40R**	Strong edge, low win rate
60%	+1R	−1R	(0.6×1) − (0.4×1) = **+0.20R**	Solid edge

> 🔑 **A 90% win rate can lose money; a 35% win rate can be highly profitable.** Stop asking "how often do I win?" and start asking "what's my expectancy?" This single reframe immunizes you against the most common beginner delusion — and against the snake-oil sellers who advertise win rates.

3.5 Profit factor — a quick health check

Profit factor = total gross profit ÷ total gross loss (in $ or R).

```
Profit Factor = Σ(winning trades) / Σ(losing trades, absolute)
```

- **PF < 1.0** → losing system.
- **PF ≈ 1.0–1.3** → marginal.
- **PF 1.3–1.7** → good, tradeable.
- **PF > 2.0** → excellent (be suspicious if a backtest shows this — possible overfitting, Ch. 4).

Profit factor and expectancy tell related stories; report both. PF is intuitive ("for every $1 I lose, I make $X"); expectancy tells you per-trade edge.

3.6 Sample size — why 10 trades tell you nothing

A positive result over a few trades is **noise**, not evidence. Variance dominates small samples.

- **Any system can win (or lose) 5–10 in a row by pure luck.** Drawing conclusions from a handful of trades is the road to ruin — you'll trust a lucky losing system or abandon a good one in an unlucky streak.
- **Rules of thumb:**

Sample size	What it tells you
< 20 trades	Essentially nothing — pure noise
30 trades	Rough hint, low confidence
100+ trades	Meaningful estimate of your edge
300+ trades	Solid confidence in the statistics

> 🔑 **The law of large numbers is your ally and your judge.** An edge only reveals itself over a large sample. This is *why* backtesting (Ch. 4) and patient journaling (Ch. 6) matter — they're how you accumulate a sample big enough to trust. Never judge a system, or yourself, on a small number of trades.

3.7 Variance, losing streaks & drawdown

Even a great positive-expectancy system has **brutal losing streaks** — this is normal and unavoidable.

- With a 40% win rate, a run of **8+ consecutive losses** is statistically expected over a few hundred trades. If that streak would break you (financially or emotionally), your *risk per trade is too high* (Vol 1).
- **Drawdown** (peak-to-trough equity decline) is the cost of admission for the edge. You must know your system's likely drawdown *in advance* so you can survive it without panicking or abandoning a working system.
- **Expectancy tells you the edge; drawdown tells you whether you can survive long enough to collect it.** Both matter.

```
The two questions every system must answer:
   1. Is expectancy positive?        → Do I have an edge at all?
   2. Can I survive the drawdown?     → Will I still be here when it pays off?
```

3.8 Putting it together — the profitability identity

```
Long-run profit  ≈  Expectancy (R)  ×  Number of trades  ×  Risk per trade ($)

To grow the account, you can:
   • Raise expectancy    → better strategy / filters / exits (Ch. 8)
   • Take more (quality) trades → more opportunities at the SAME edge
   • Increase risk per trade    → ⚠ amplifies drawdown too; change LAST and SLOWLY

Survival constraint: risk per trade must be small enough to outlast the
worst expected losing streak. Edge means nothing if you blow up first.
```

> 🔑 The professional path is: **establish a positive expectancy → execute it over a large sample with survivable risk → only then scale.** Most blow-ups come from inverting this — scaling risk before proving the edge. Get the math right and trading becomes what it should be: the patient harvesting of a measured edge.

3.9 Chapter summary

```
• EDGE = positive average outcome per trade after costs (statistical, not predictive).
• Work in R-MULTIPLES (1R = amount risked).
• EXPECTANCY = (Win% × AvgWinR) − (Loss% × AvgLossR). Positive = an edge. THE key number.
• Win rate alone is meaningless: 90% can lose, 35% can win. It's win% AND reward:risk.
• PROFIT FACTOR = gross profit / gross loss (1.3–1.7 good; >2 suspect overfitting).
• SAMPLE SIZE: need 100+ trades for meaningful stats; <20 is pure noise.
• Positive-expectancy systems still have long losing streaks & drawdowns — size to survive them.
• Profit ≈ expectancy × #trades × risk. Prove the edge first; scale risk last.
```

CHAPTER 4

Backtesting Properly

Backtesting is applying your strategy's rules to *historical* data to estimate its edge before risking money. Done well, it's the cheapest way to learn whether a strategy is worth trading. Done badly — which is how *most* people do it — it produces beautiful, completely worthless results that lead straight to live losses. This chapter teaches the discipline that makes backtesting honest.

> ⚠️ **The central warning of this volume:** it is trivially easy to produce a backtest that looks amazing and predicts nothing. The market's history is full of patterns that existed by chance and never repeat. Your enemy in backtesting is **not the market — it's your own ability to fool yourself.** Everything below is about defeating that.

4.1 What backtesting can and can't do

It CAN:
- Estimate expectancy, win rate, profit factor, and drawdown over a large sample.
- Quickly reject bad strategies (saving real money).
- Build your familiarity with how a strategy behaves.

It CAN'T:
- Guarantee future results — *past performance does not predict the future*.
- Capture your real-world psychology, slippage, or execution mistakes.
- Save a strategy with no logical edge from eventually failing.

> 🔑 Treat a backtest as **evidence, not proof.** A good backtest earns a strategy the right to be *forward-tested* (Ch. 5) — never the right to be trusted with size immediately.

4.2 Two ways to backtest

Manual backtesting (start here)

Scroll back through historical charts (bar-by-bar replay, e.g. TradingView's replay or MT4's strategy tester in visual mode), find every instance of your setup, and **log each hypothetical trade** (entry, stop, target, result in R).
- ✅ Builds deep intuition and chart-reading skill; works for discretionary strategies; no coding.
- ❌ Slow; vulnerable to **hindsight bias** (it's tempting to "see" trades you'd have missed live).

Automated backtesting

Code the rules and run them over historical data (MT4/MT5 Expert Advisors, Python with libraries, TradingView Pine Script strategies, dedicated platforms).
- ✅ Fast, tests thousands of trades, perfectly consistent, only works for fully mechanical rules.
- ❌ Requires coding; **far easier to curve-fit**; only as good as your data and assumptions.

Most retail traders should start manual to build judgment, then automate the mechanical parts once the logic is proven.

4.3 The seven deadly biases (how backtests lie to you)

These are the ways a backtest produces fake results. Know each one by name so you can hunt it down.

1. Curve-fitting / Overfitting ⭐ (the big one)

Tuning rules and parameters until they fit the historical data *perfectly* — capturing **noise**, not edge. A curve-fit system looks flawless on the past and falls apart live.
- **Tells:** many parameters, oddly specific values (e.g. "RSI 27.5", "exit after 13 bars"), incredible equity curve, performance collapses if you nudge any setting.
- **Defense:** few parameters, logical rules, and **out-of-sample testing** (4.4).

2. Look-ahead bias

Using information that wouldn't have been available at decision time (e.g. using a candle's close to enter *at* that close, or referencing future bars). Inflates results impossibly.
- **Defense:** only act on data that was knowable *before* the decision point.

3. Hindsight bias (manual testing's curse)

Unconsciously taking the trades you *know* worked and skipping the ambiguous ones. Your rules feel clearer in hindsight than they were live.
- **Defense:** mechanical, pre-written rules; use bar-replay so you can't see the future; log *every* qualifying setup, including the losers and the ugly ones.

4. Survivorship bias

Testing only on instruments/periods that "survived" or trended nicely, ignoring the ones that died or chopped. Common when cherry-picking a pair that happened to trend.
- **Defense:** test across multiple pairs and varied market regimes.

5. Optimization / data-snooping bias

Trying hundreds of variations and picking the best — by chance, *something* will look great on past data. You've selected for luck.
- **Defense:** out-of-sample/walk-forward validation; limit how much you optimize.

6. Ignoring costs

Backtesting without spread, commission, and slippage. A strategy with small average wins can be profitable on paper and a loser after real costs (Vol 1).
- **Defense:** subtract realistic spread/commission/slippage from every simulated trade.

7. Unrealistic execution

Assuming perfect fills at exact prices, no requotes, trading illiquid hours, or sizes the market couldn't absorb.
- **Defense:** model conservative, realistic fills; assume you get the *worse* side of ambiguity.

4.4 In-sample vs out-of-sample — the honesty test ⭐

The most important technique for trustworthy backtesting: **split your data.**

```
┌───────────────────────────────────┬───────────────────────────────┐
│   IN-SAMPLE (e.g. 70%)            │   OUT-OF-SAMPLE (e.g. 30%)    │
│   Build & refine the strategy│   NEVER touched during dev    │
│   here. Optimize here.            │   Used ONCE to validate.      │
└───────────────────────────────────┴───────────────────────────────┘
```

- Develop and tune the strategy **only** on the in-sample data.
- Then run it **once** on the untouched out-of-sample data. If performance **holds up**, the edge is likely real. If it **collapses**, you curve-fit — back to the drawing board.
- **Walk-forward analysis** extends this: repeatedly optimize on a window, test on the next unseen window, and roll forward — simulating how the strategy would have adapted over time. It's the gold standard for mechanical systems.

> 🔑 **If a strategy only works on the data you built it on, it doesn't work.** Out-of-sample validation is the difference between discovering an edge and memorizing the past. Protect your out-of-sample data like gold — you only get to use it honestly once.

4.5 A disciplined backtesting procedure

```
1. SPECIFY    Write the exact rules first (Ch. 2). Do NOT change them mid-test.
2. SPLIT      Reserve out-of-sample data you won't look at during development.
3. SAMPLE     Test enough trades (100+) across varied conditions & multiple pairs.
4. LOG        Record EVERY qualifying trade in R — winners, losers, and ugly ones.
              (Use the backtest log template, Ch. 9.)
5. COST       Subtract realistic spread/commission/slippage from each trade.
6. MEASURE    Compute expectancy, win%, profit factor, max drawdown (Ch. 7).
7. VALIDATE   Run ONCE on out-of-sample data. Does the edge survive?
8. DECIDE     Meets your pre-set criteria (Ch. 2)? → forward-test. If not → revise or kill.
```

4.6 Interpreting results with healthy skepticism

- **Too good to be true usually is.** A 90% win rate or PF > 3 in backtest screams overfitting or a bug (often look-ahead bias). Real edges are modest.
- **Look at the equity curve, not just the bottom line.** A smooth, steady rise is more trustworthy than a jagged line that made all its money in one freak period. Check the **drawdowns** — could you stomach the worst one *live*?
- **Check trade distribution.** Is the profit spread across many trades, or dependent on a few outliers? Outlier-dependent results are fragile.
- **Stress-test:** does it still work if you shift parameters slightly, test a different period, or another pair? Robust edges degrade *gracefully*; fitted ones break.

> 🔑 The right attitude is **active distrust of your own backtest.** Assume it's flawed and try to break it. The strategies that survive your honest attempts to disprove them are the ones worth real money.

4.7 Chapter summary

```
• Backtesting = applying rules to history to estimate edge. Evidence, not proof.
• Manual builds intuition (start here); automated is fast but easy to curve-fit.
• Beware the 7 biases: OVERFITTING, look-ahead, hindsight, survivorship,
  optimization/data-snooping, ignored costs, unrealistic execution.
• OUT-OF-SAMPLE validation is the honesty test: develop on in-sample, validate ONCE
  on untouched data. Walk-forward is the gold standard. If it only works where you
  built it, it doesn't work.
• Always include realistic costs. Demand 100+ trades across varied conditions.
• Distrust beautiful results; inspect the equity curve, drawdowns, and trade distribution.
```

A backtest that survives this discipline has earned the right to be tested *forward* — on data the strategy has never seen, in conditions closer to real trading. That's Chapter 5.

CHAPTER 5

Forward Testing: Demo to Live

A backtest tells you how a strategy *would have* done on the past. **Forward testing** tells you how it does on data it has never seen — in real time, with real execution friction, and (eventually) real emotion. It's the bridge between a promising backtest and risking actual capital, and skipping it is one of the most expensive shortcuts a trader can take.

5.1 Why forward testing is non-negotiable

Backtesting has an unavoidable blind spot: **the strategy was developed with knowledge of the period it's tested on.** Even with out-of-sample discipline, you chose the rules in a world where that history already happened. Forward testing removes that contamination entirely — the future genuinely hasn't happened yet.

It also surfaces things a backtest *can't*:
- **Real execution:** actual spreads, slippage, fills, requotes, platform quirks.
- **Real timing:** can you actually be at the screen when setups occur?
- **Real psychology** (once live): hesitation, fear, the urge to override rules.
- **Regime change:** does the edge still exist in *today's* market, not just history's?

> 🔑 Backtest answers *"did this edge exist?"* Forward test answers *"does this edge exist NOW, and can I actually execute it?"* You need both yeses before sizing up.

5.2 The validation ladder — climb it in order

Move to the next rung only after the current one passes. Each rung adds realism and risk gradually.

```
RUNG 1 — BACKTEST           Historical edge established (Ch. 4). Pre-set criteria met.
            ↓
RUNG 2 — PAPER / DEMO        Trade the LIVE market in real time with virtual money.
            ↓                Validates execution & timing without financial risk.
RUNG 3 — LIVE, MICRO          Real money, smallest size, 0.25–0.5% risk. Adds real
            ↓                 emotion — the only thing demo can't replicate.
RUNG 4 — LIVE, NORMAL          Scale toward your normal (still ≤1%) risk, slowly,
                              ONLY after a proven, journaled live track record.
```

Most blow-ups come from jumping rungs — going from a pretty backtest straight to meaningful live size, skipping the two rungs that catch execution and psychology problems.

5.3 Demo / paper trading — do it right

Demo is free and risk-free, which is its strength *and* its weakness — it's easy to treat it carelessly and learn nothing.

Make demo meaningful:
- **Trade it exactly like real money.** Use realistic position sizes and your real risk %. Don't trade 50 lots "because it's fake" — you'll build worthless habits.

- **Follow the strategy spec mechanically.** The point is to validate *the system*, not to freestyle.
- **Journal every trade** (Chapter 6) exactly as you will live.
- **Run a meaningful sample** — weeks of trading and ideally 30–50+ trades before judging.

What demo proves: that you can execute the strategy in real time, that your rules are actually clear enough to follow live, and that the edge appears to persist in current conditions.

Demo's honest limitation: it **cannot replicate the emotion** of real money. People are disciplined on demo and fall apart live. That's exactly why Rung 3 (live micro) exists — and why you start it *tiny*.

5.4 The leap to live — managing the psychology gap

The single biggest surprise traders report: **the same strategy feels completely different with real money.** Fear makes you exit winners early and hesitate on valid setups; the sting of real losses tempts you to override your stop.

How to cross the gap:
- **Start so small the money barely matters** (micro lots, 0.25–0.5% risk). The goal of Rung 3 is **not profit — it's executing your proven system under real emotion.**
- **Treat the rules as sacred.** The whole point is to prove you can run the *tested* system, not a panicked variation of it.
- **Compare live results to your backtest/demo.** A reasonable match suggests the edge is real and you're executing well. A big negative gap usually means an *execution/psychology* problem (you're not trading your system), not a strategy problem — the journal will reveal which.

> 🔑 If your live results diverge sharply from a well-validated backtest, suspect **yourself before the strategy.** The most common cause is deviation from the rules under emotional pressure — which is a journaling and discipline fix (Ch. 6, and Vol 2), not a reason to scrap the system.

5.5 How long, and how many trades?

There's no magic number, but useful guidance:
- **Demo:** at least a few weeks and ~30+ trades, enough to confirm you can execute and the edge shows up.
- **Live micro:** long enough to experience real winning *and* losing streaks and prove you hold discipline through both — often 1–3 months and 30–50+ trades.
- **The deciding factor isn't time — it's a journaled track record** showing positive expectancy *and* consistent rule-following across a real sample (Ch. 7).

Be patient. The cost of a few extra months of testing is trivial; the cost of going live too early with too much size is how accounts die.

5.6 When to advance, hold, or go back

```
ADVANCE a rung when:  expectancy is positive over a real sample AND you followed
                      your rules consistently AND you can stomach the drawdowns.

HOLD / repeat when:   results are inconclusive, the sample is too small, or your
                      rule-following is shaky. More data, same rung.
```

```
GO BACK when:         the edge clearly fails out-of-sample/live, OR you keep
                      breaking rules (fix discipline before risking more).
```

⚠️ **Never "promote" a strategy out of impatience or because you're bored.** Promotion is earned by *data*. The market will still be there next month; an account blown by rushing may not be.

5.7 Chapter summary

```
• Forward testing validates a strategy on data it has NEVER seen — removing the
  contamination backtests can't avoid, and exposing execution/timing/psychology.
• Climb the VALIDATION LADDER in order: Backtest → Demo → Live-Micro → Live-Normal.
  Don't skip rungs.
• Demo only works if you trade it like real money, by the spec, fully journaled.
• Demo CAN'T replicate real-money emotion — that's why live-micro (tiny size) exists.
• If live diverges from a solid backtest, suspect YOUR execution first, not the edge.
• Advance rungs on DATA (positive expectancy + rule-following over a real sample),
  never on impatience.
```

Every rung of that ladder depends on one thing: a disciplined record of what you actually did. That record — the journal — is the engine of the entire development loop, and it's the subject of Chapter 6.

CHAPTER 6

The Journaling System

The journal is the **engine of improvement** for the entire series. Volume 1 gave you a starter template; Volume 2 showed that every great trader keeps detailed records; this chapter turns journaling into a **system** — a structured database of your trading that powers the analytics (Chapter 7) and the refinement loop (Chapter 8). Without it, you are flying blind, and no amount of strategy knowledge can save a trader who doesn't measure.

> 🔑 **You cannot improve what you do not measure.** The journal converts scattered experience into structured data, and structured data into insight. It is the difference between *ten years of experience* and *one year of experience repeated ten times*.

6.1 Why most traders don't journal (and why that's their downfall)

Journaling is tedious, it's confronting (it forces you to face losses and mistakes honestly), and its payoff is delayed. So most traders skip it — and then wonder why they never improve. They remember their wins, forget their losses, misjudge their own win rate, and "fix" the wrong things.

A journaling *system* defeats this by making logging **routine, structured, and non-optional** — part of every trade, like the stop-loss.

6.2 What to log — the anatomy of a journal entry

A useful journal captures three layers: **the plan, the result, and the process.** The process layer is what most people omit and what matters most.

```
── IDENTITY ──────────────────────────────────────────────
  Trade #, Date/Time, Pair, Session, Strategy/Setup name, Long/Short

── THE PLAN (recorded at entry) ──────────────────────────
  Entry price, Stop, Target(s), Position size, Risk %, Stop distance (pips),
  Planned R:R, The CONFLUENCE / reasons for the trade

── THE RESULT ────────────────────────────────────────────
  Exit price, Outcome (win/loss/BE), Pips, $ P/L, R-multiple,
  Costs (spread/commission/swap)

── THE PROCESS (the highest-value part) ──────────────────
  □ Did I follow my plan EXACTLY?  (Yes / No — and what I broke)
  □ Entry & exit execution quality
  □ Emotional state (calm / FOMO / fearful / greedy / revenge / bored)
  □ Mistakes made
  □ What I did WELL
  □ Lesson / one thing to do differently
  □ Screenshot (chart at entry & exit)

── CONTEXT (optional but powerful for analytics) ─────────
  Market regime (trend/range), day of week, time of day, MAE & MFE (Ch. 7)
```

> 🔑 The **"Did I follow my plan?"** field is the most important in the whole journal. It lets you separate *strategy* problems (followed the plan, still lost over a large sample → fix the strategy) from *discipline* problems (broke the plan → fix yourself). These require opposite responses, and only the journal can tell them apart.

6.3 The tools — pick one and start

There's no single right tool; the best one is the one you'll actually use consistently.

Tool	Best for	Notes
Spreadsheet (Excel / Google Sheets)	Most traders	Flexible, free, auto-calculates KPIs with formulas. **Recommended starting point.**
Markdown / Notion / docs	Narrative + screenshots	Great for qualitative review; weaker at analytics. (Vol 1's template is this style.)
Dedicated journal apps	Automation & deep analytics	Auto-import trades, rich stats/charts. Convenient; costs money; less customizable.
Trading platform reports	Raw trade history	A data *source* to feed your journal — not a substitute for the process layer.

Recommendation: start with a **spreadsheet** — one row per trade, columns matching the fields above. It costs nothing, computes your metrics automatically (Chapter 7), and teaches you exactly what drives your results. (Schema provided in Chapter 9.)

6.4 A spreadsheet schema that auto-computes your edge

One row per trade; let formulas do the analytics:

```
Core columns:
  Date | Pair | Setup | Dir | Entry | Stop | Target | Exit | Size | Risk$ |
  R-multiple | Win/Loss | Followed-Plan? | Emotion | Mistake | Note | Screenshot-link

Derived (formulas across the rows):
  • Win rate        = COUNT(wins) / COUNT(trades)
  • Average win (R) / Average loss (R)
  • EXPECTANCY (R)  = (Win% × AvgWinR) − (Loss% × AvgLossR)
  • Profit factor   = SUM(win R) / ABS(SUM(loss R))
  • Cumulative R    → plot as your EQUITY CURVE
  • % plan-followed = COUNT(Followed=Yes) / COUNT(trades)
```

Now your journal isn't just a diary — it's a **live dashboard of your edge** that updates with every trade. This is the bridge to Chapter 7.

6.5 The review cadence — journaling is nothing without review

Logging is only half the system; **review** is where the value is extracted. Build a fixed rhythm:

```
PER TRADE (immediately after closing):
  • Fill the full entry while it's fresh — especially emotion & mistakes.
```

```
DAILY (end of session, ~10 min):
  • Did I follow my rules today? Any tilt? Quick note of lessons.

WEEKLY (~30–45 min):
  • Compute the week's stats (win%, R, expectancy, profit factor).
  • % of trades I followed the plan.
  • Best decision & worst mistake of the week.
  • One adjustment for next week.

MONTHLY (deep review):
  • Full KPI review (Ch. 7): equity curve, drawdown, performance BY setup,
    BY session, BY day, BY emotion.
  • Is each strategy's edge holding? Refine ONE variable (Ch. 8).

PERIODIC (quarterly):
  • Big-picture: which setups to keep, cut, or scale? Are my rules still valid?
```

🔑 **Grade yourself on PROCESS, not P/L** (Vol 2, Ch. 6). The most important review question isn't "did I make money?" — it's *"did I follow my system?"* Over a large sample, a faithfully executed positive-expectancy system *must* make money; your job in review is to ensure faithful execution and to feed the refinement loop.

6.6 Turning journal insights into action

The journal exists to drive *change*. Common, high-value patterns it reveals:

- *"My expectancy is positive overall but deeply negative on counter-trend trades."* → stop taking counter-trend setups.
- *"I lose most when I trade during the Asian session / around news."* → add a condition filter (Ch. 2).
- *"My 'broke the plan' trades have a far worse R than my disciplined ones."* → it's a discipline problem; tighten execution, not the strategy.
- *"Win rate is fine but I cut winners early (low avg win R)."* → fix the *exit* (let winners run; Vol 3).
- *"All my profit comes from 2 outlier trades."* → fragile edge; investigate before scaling.

Each insight becomes a **specific, testable change** — fed back into the development loop (Ch. 8), validated, and re-measured. That loop, powered by the journal, is how trading skill actually compounds.

6.7 Chapter summary

```
• The journal is the ENGINE of improvement — you can't improve what you don't measure.
• Log three layers: the PLAN, the RESULT (in R), and the PROCESS (emotion, mistakes,
  and especially "Did I follow my plan?").
• The "followed-plan?" field separates STRATEGY problems from DISCIPLINE problems —
  which need opposite fixes.
• Start with a SPREADSHEET (one row/trade) that auto-computes your KPIs & equity curve.
• Journaling without REVIEW is useless: build a per-trade / daily / weekly / monthly cadence.
• Grade yourself on PROCESS, not P/L. Turn each insight into one specific, testable change.
```

Your journal is now generating data. Chapter 7 turns that data into the specific **metrics** that tell you whether — and *why* — your trading works.

CHAPTER 7

Performance Metrics & Analytics

Your journal (Chapter 6) is now a database of trades. This chapter is the **analytics layer** — the specific metrics (KPIs) that turn that data into a precise diagnosis of your trading: not just *whether* you're profitable, but *why*, and *where the leaks are*. These are the numbers professionals live by.

7.1 The core profitability KPIs

These five answer "do I have an edge, and how strong?" (foundations from Chapter 3):

KPI	Formula	What it tells you	Healthy range
Win rate	wins ÷ total	How often you win (context only!)	Any — judge with R
Avg win / Avg loss (R)	mean win R, mean loss R	The *size* side of the edge	Avg win > avg loss ideal
Expectancy (R) ⭐	(Win%×AvgWinR) − (Loss%×AvgLossR)	**Average profit per trade — the edge**	> 0; higher better
Profit factor	gross profit ÷ gross loss	$ made per $ lost	1.3–1.7 good; >2 suspect
R-expectancy × #trades	total R captured	Total edge expressed	Trending up

🔑 Report **expectancy and profit factor together**, always alongside **sample size**. A great expectancy over 12 trades means nothing (Ch. 3). The number of trades is the confidence level on every other metric.

7.2 Risk & survival KPIs

Profitability tells you the upside; these tell you whether you'll *survive* to collect it.

Maximum drawdown ⭐

The largest peak-to-trough fall in your equity, in % (or R). **The single most important risk metric** — it's the worst pain the system put you through, and a preview of what you must be able to endure live.
- Know it *before* going live so a normal drawdown doesn't make you abandon a working system.
- Recovery math is brutal (Vol 1): −50% needs +100% to recover. Keeping max drawdown modest is survival.

Longest losing streak

The most consecutive losers. Tells you the emotional and financial endurance the system demands. If the streak would break you at your risk %, **lower the risk %.**

Risk of ruin

The probability of losing enough to be unable to continue, given your edge, win rate, and risk per trade. The practical takeaway: **small risk per trade (≤1%) + positive expectancy = risk of ruin near zero.** Large risk per trade can produce meaningful ruin probability *even with a positive edge.*

Return / drawdown ratio

Net return ÷ max drawdown — "how much pain per unit of gain." Higher is better; it captures *risk-adjusted* performance better than raw return alone.

7.3 The equity curve — your trading at a glance ⭐

Plot **cumulative R** (or account balance) trade by trade. The *shape* tells you more than any single number:

```
Healthy edge:                      Fragile / problematic:
  R                                  R
  |            /                     |         /\    /
  |        /\ /                      |        /  \  /
  |      /\/  \/                     |    /\ /    \/
  |   /\/                            |   /  \/      (one spike, then chop/decline)
  | /                                | /
  └─────────────── trades            └─────────────── trades
   Steady rise, shallow dips          Lumpy, outlier-dependent, deep drawdowns
```

- **Smooth, steadily rising** → a robust, consistent edge.
- **One big jump then flat/declining** → profit depends on outliers; fragile.
- **Deep, prolonged drawdowns** → high risk; check sizing and the strategy's regime fit.
- A rising equity curve with a **flat or rising "% plan-followed"** line is the signature of a maturing trader.

7.4 Segmentation analytics — where the edge really lives

This is where the journal earns its keep. **Slice your results by category** to find your real strengths and the hidden leaks. A net-profitable trader is often *very* profitable in some buckets and bleeding in others.

```
Break your trades down BY:
  • SETUP / strategy    → which edges actually work? (cut the losers)
  • SESSION / time      → are you profitable in London but losing in Asia?
  • DAY OF WEEK         → any consistently bad day?
  • PAIR                → is one instrument dragging you down?
  • DIRECTION           → are your shorts much worse than your longs?
  • MARKET REGIME       → trend vs range — does your edge need one?
  • EMOTIONAL STATE     → how much worse are "FOMO"/"revenge" trades?
  • FOLLOWED-PLAN y/n   → the expectancy gap between disciplined & undisciplined trades
```

> 🔑 **The most common discovery:** "I'm net positive, but one or two buckets are killing me." Cutting your worst-performing setup, session, or pair often does more for your bottom line than finding a brand-new strategy. **Subtraction is an underrated edge.**

7.5 Trade-quality analytics: MAE & MFE

Two advanced metrics that diagnose your **entries and exits** specifically:

- **MAE (Maximum Adverse Excursion):** the furthest a trade went *against* you before you closed it. Aggregated, it reveals whether your **stops** are well-placed — e.g. if winners rarely go more than 0.5R against you, a 1R stop may be wider than necessary.
- **MFE (Maximum Favorable Excursion):** the furthest a trade went *in your favor* before you closed it. Reveals whether you're **leaving profit on the table** — e.g. if trades routinely reach +3R but you exit at +1R, your targets/trailing need work (you're cutting winners short).

Together, MAE/MFE turn vague feelings ("I think I exit too early") into evidence you can act on.

7.6 System Quality — combining edge and consistency

For a single summary of system quality, some traders use metrics that blend **expectancy with consistency** (how reliable the per-trade results are, via their variability) and **sample size**. The intuition you need, without the formula worship:

- A system with a **steady** +0.3R per trade is often *better to trade* than one averaging +0.5R with wild swings — because the steadier one has shallower drawdowns and is far easier to execute with discipline.
- **Consistency has real value.** When comparing strategies, don't just rank by average return; weigh the **smoothness of the equity curve and the depth of drawdowns** too.

> 🔑 The best system isn't the one with the highest expectancy on paper — it's the one with a positive edge you can **actually execute and survive**, given its drawdowns and your psychology.

7.7 Reading the numbers honestly

- **Always anchor to sample size.** Every KPI is an estimate; small samples give noisy, untrustworthy estimates.
- **Look for stability over time**, not a single great month. Recent performance vs overall — is the edge persisting or decaying?
- **Beware over-reaction.** A normal losing streak is not a broken system (Ch. 3). Distinguish variance from genuine edge decay (a sustained, large divergence over a meaningful sample).
- **Let the data, not emotion, drive decisions** — but interpret it with the context of how trading actually works.

7.8 Chapter summary

```
• Core KPIs: win rate (context only), avg win/loss R, EXPECTANCY, profit factor — always
  with SAMPLE SIZE as the confidence level.
• Survival KPIs: MAX DRAWDOWN (the key risk metric), longest losing streak, risk of ruin,
  return/drawdown ratio. Know your drawdown BEFORE going live.
• The EQUITY CURVE's shape reveals robustness vs fragility at a glance.
• SEGMENT results by setup/session/day/pair/direction/regime/emotion/followed-plan —
  cutting your worst bucket often beats finding a new strategy.
• MAE/MFE diagnose stop placement and whether you're cutting winners short.
• Prefer a steady, survivable edge over a high-but-wild one. Judge risk-adjusted, not raw.
```

You can now diagnose your trading precisely. Chapter 8 uses that diagnosis to **optimize and iterate** — and to size positions so the edge compounds without blowing up.

CHAPTER 8

Optimizing, Iterating & Position Sizing

You have a defined strategy, a validation process, a journal, and analytics. This chapter closes the development loop: how to **refine** a system using data without breaking it, when to **keep or kill** a strategy, and how **position sizing** turns a proven edge into compounding growth — or into a blow-up if done wrong.

8.1 The refinement loop — improve without curve-fitting

Optimization done carelessly is just curve-fitting on your own live data (Chapter 4's sin, repeated). Done well, it's disciplined, evidence-based improvement.

```
THE GOLDEN RULES OF ITERATION
1. Change ONE variable at a time. Multiple simultaneous changes make it impossible
   to know what helped or hurt.
2. Base every change on JOURNAL EVIDENCE over a meaningful sample — never on a
   few recent trades or a gut feeling after a loss.
3. RE-VALIDATE each change (ideally on out-of-sample / forward data) before trusting it.
4. Prefer SUBTRACTION: removing a losing bucket (Ch. 7) usually beats adding rules.
5. Stay SIMPLE. If "improving" means adding parameters, be suspicious — you may be
   fitting noise, not strengthening the edge.
```

🔑 **The paradox of optimization:** the more you tune a system to its past results, the *worse* it often performs in the future. Your goal is a **robust** edge that survives changing conditions — not a fragile one perfected for history. Resist the urge to over-engineer.

8.2 What's worth optimizing (in priority order)

Use your segmentation analytics (Ch. 7) to target the highest-leverage changes:

1. **Cut what loses.** If a setup, session, pair, or direction has negative expectancy over a real sample, *stop trading it*. Pure subtraction, biggest impact, lowest risk.
2. **Fix the exits.** MAE/MFE often show winners cut short or stops poorly placed. Improving exits (let winners run, refine targets/trailing) raises avg-win-R — usually the fastest expectancy gain.
3. **Tighten the condition filter.** Add a *logical* regime/time/news filter that removes a cluster of losers (e.g. "don't trade this in ranges / around red news").
4. **Refine entries last.** Entry tweaks have the smallest, noisiest impact and the highest curve-fitting risk. Touch them least.

🔑 Notice the order: **exits and filters before entries.** Beginners do the reverse, endlessly fiddling with entry signals while ignoring the exit and risk decisions that actually move the needle.

8.3 When to keep, fix, or kill a strategy

Distinguish **variance** (normal, expected bad runs) from genuine **edge decay** (the strategy stopped working). Getting this wrong in either direction is costly — abandoning a good system in a normal drawdown, or clinging to a dead one.

```
KEEP & TRADE   → Positive expectancy over a large sample; current drawdown is within
                 historically normal range; you're following the rules.
FIX / REFINE   → Net positive but a clear bucket is leaking (Ch. 7) → cut/adjust that
                 bucket; or a discipline gap (followed-plan analytics) → fix execution.
KILL           → Edge fails out-of-sample/forward; expectancy is negative over a LARGE
                 sample of RULE-FOLLOWED trades; or the market regime it needs is gone
                 and not returning. Don't marry a strategy.
```

The discipline-vs-strategy test (from Ch. 6): before killing a strategy, check the **"followed-plan?"** analytics. If your *rule-following* trades are profitable and only your *rule-breaking* trades lose, the strategy is fine — **you** are the problem to fix (Vol 2's psychology), not the system.

8.4 Position sizing models — converting edge into growth

A positive edge only builds wealth through **position sizing**. The model you choose governs how fast you grow and how deep your drawdowns run.

Model	How it works	Pros / Cons
Fixed lot	Same lot size every trade	Simple; but risk varies with stop distance & ignores account growth. Weak.
Fixed fractional (% risk) ⭐	Risk a fixed **%** of equity per trade (e.g. 1%)	**The standard.** Risk scales with the account, auto-compounds, survivable. *Use this.*
Fixed ratio	Increase size after each fixed profit increment	Smoother scaling for growing accounts; more complex.
Kelly criterion	Mathematically "optimal" % from your edge & odds	Maximizes *theoretical* growth — but is **far too aggressive** in practice (huge drawdowns, and it assumes you know your true edge exactly).

Fixed-fractional is the answer for almost everyone

Risk a constant small % of *current equity* per trade (Vol 1's method). As the account grows, position sizes grow automatically (compounding); as it shrinks, they shrink automatically (capital preservation). It's simple, robust, and mathematically sound.

On Kelly — useful idea, dangerous in full

The Kelly criterion gives the growth-optimal fraction, but **full Kelly produces gut-wrenching drawdowns and assumes you know your edge precisely (you don't).** Practitioners who use it at all use a small fraction — *"half-Kelly"* or less. **The practical lesson:** there's a mathematically optimal aggressiveness, and you should stay *well below* it. When unsure, **risk less.** A smaller, survivable edge compounded for years beats an "optimal" one that blows up in a bad streak.

> 🔑 **Sizing decides survival.** The same positive-expectancy system can compound beautifully at 1% risk or blow up at 10% risk — *identical edge, opposite outcomes.* Edge earns the money; sizing decides whether you live to keep it.

8.5 Scaling up — earning the right to trade bigger

When your validated, journaled track record justifies it, scale **gradually**:

```
• Keep risk as a FIXED %; let compounding grow the absolute size automatically.
• Increase the % only in small steps (e.g. 0.5% → 0.75% → 1%), and only after a
  sustained, journaled record of positive expectancy AND rule-following.
• Re-check drawdown tolerance at the larger size — can you stomach it in real money?
• Scale DOWN when out of rhythm or in drawdown (Vol 2's pros do exactly this).
• Never jump risk to chase a hot streak — that's when most blow-ups happen.
```

8.6 The continuous-improvement mindset

Markets evolve; edges decay; you improve. Trading is never "solved" — it's a permanent loop of **hypothesize → test → execute → measure → refine.**

- Maintain a **strategy pipeline**: as one edge weakens, you should be researching and validating the next, using this volume's process.
- Keep a **research log** of ideas to test (separate from your live journal).
- Review your whole *system* periodically (quarterly), not just individual trades.
- Accept that **iteration never ends** — the traders who last are perpetual students running a disciplined process, not people who found a magic setup.

> 🔑 The edge that matters most isn't any single strategy — it's your **mastery of the development loop itself.** Strategies come and go; a trader who can reliably *find, validate, and refine* edges has the only durable advantage in a changing market.

8.7 Chapter summary

```
• Iterate carefully: ONE variable at a time, on journal EVIDENCE, RE-VALIDATED,
  favoring SUBTRACTION and SIMPLICITY. Over-tuning fits noise and hurts the future.
• Optimize in order: CUT losers → fix EXITS → tighten FILTERS → entries LAST.
• Keep / Fix / Kill by distinguishing variance from edge decay — and check the
  "followed-plan?" analytics first (discipline problem vs strategy problem).
• Position sizing: use FIXED-FRACTIONAL (% risk). Kelly is theoretically optimal but
  far too aggressive — stay well below it; when unsure, risk LESS.
• Same edge + different sizing = compounding vs blow-up. Sizing decides survival.
• Scale risk % up only on a proven record, in small steps; scale down when struggling.
• The durable edge is mastery of the LOOP itself: hypothesize → test → execute →
  measure → refine, forever.
```

You now have the complete development system. The final section is a **pack of ready-to-use templates** to run all of it.

APPENDIX

Templates & Systems Pack

A. Strategy Specification Sheet

> One per strategy. If a stranger couldn't trade identically from this sheet, it isn't finished.

```
STRATEGY NAME: ________________________________   Version: ____  Date: ______

1. HYPOTHESIS / EDGE (one sentence):
   _____________________________________________________________________
   Logical reason this edge should exist: ______________________________

2. MARKET(S):        ______________________  (why these: __________________)
3. TIMEFRAMES:       Bias ____  | Setup ____  | Entry ____
4. CONDITION FILTER (when VALID / when to STAND ASIDE):
   • Valid when:   __________________________________________________
   • Invalid when: __________________________________________________

5. ENTRY RULES (exact, repeatable — all must be TRUE):
   1) ______________________________________________________________
   2) ______________________________________________________________
   3) ______________________________________________________________
   Entry execution: □ on close  □ on retest  □ limit at level

6. STOP-LOSS RULE (the invalidation point): ______________________________
7. TAKE-PROFIT RULE (targets / trailing / scaling): ______________________
8. MINIMUM RISK:REWARD: 1 : ____

9. RISK PER TRADE: ____ %    Position-size formula: (Acct × Risk%) ÷ (Stop pips × pip value)
10. TRADE MANAGEMENT: break-even at ____ R | partial at ____ | max hold: ____
11. WHAT INVALIDATES THE STRATEGY (stop trading it when): ________________

ACCEPTANCE CRITERIA (set BEFORE testing — Ch. 2 & 4):
   □ Expectancy > ____ R   □ Profit factor ≥ ____   □ Max drawdown ≤ ____ %
   □ Min sample ≥ ____ trades   □ Holds OUT-OF-SAMPLE
```

B. Backtest Log

> One row per simulated trade. Costs included. Log winners, losers, and ugly ones.

```
| # | Date | Pair | Dir | Entry | Stop | Target | Exit | Stop(pips) | R | W/L | Cost | Notes |
|---|------|------|-----|-------|------|--------|------|------------|---|-----|------|-------|
| 1 |      |      |     |       |      |        |      |            |   |     |      |       |
| 2 |      |      |     |       |      |        |      |            |   |     |      |       |

SUMMARY (auto-computed):
  Sample size: ____   In-sample: ____  Out-of-sample: ____
  Win rate: ____%   Avg win: ____R   Avg loss: ____R
  EXPECTANCY: ____R   Profit factor: ____   Max drawdown: ____%
  Verdict vs acceptance criteria:  □ PASS → forward-test   □ FAIL → revise/kill
```

C. Trade Journal Schema (spreadsheet — one row per LIVE/demo trade)

```
IDENTITY:  Trade# | Date | Time | Pair | Session | Setup | Dir
PLAN:      Entry | Stop | Target | Size | Risk% | StopPips | Planned-R:R | Confluence/Reasons
RESULT:    Exit | Outcome(W/L/BE) | Pips | $P/L | R-multiple | Costs
PROCESS:   Followed-Plan?(Y/N) | What-I-broke | Emotion | Mistake | Did-Well | Lesson | Screenshot
CONTEXT:   Regime(trend/range) | DayOfWeek | MAE(R) | MFE(R)

AUTO-CALCULATED DASHBOARD (formulas over all rows):
  Win rate · Avg win R · Avg loss R · EXPECTANCY (R) · Profit factor ·
  Cumulative R (→ equity curve chart) · Max drawdown · % plan-followed
```

D. KPI Dashboard (review at a glance — Ch. 7)

```
┌─ EDGE ──────────────────────────────────────────────────────────┐
│ Sample size: ____    Win rate: ____%    Expectancy: ____R         │
│ Avg win: ____R   Avg loss: ____R   Profit factor: ____            │
├─ SURVIVAL ──────────────────────────────────────────────────────┤
│ Max drawdown: ____%   Longest losing streak: ____                 │
│ Return/Drawdown: ____   Current drawdown: ____% (normal? Y/N)      │
├─ EQUITY CURVE ──────────────────────────────────────────────────┤
│ Shape: ◻ steady-rise  ◻ outlier-dependent  ◻ choppy/declining     │
├─ SEGMENTATION (expectancy by bucket - find leaks) ──────────────┤
│ Best setup: ______ (__R)     Worst setup: ______ (__R)            │
│ Best session: ____ (__R)     Worst session: ____ (__R)            │
│ Longs: __R  Shorts: __R   Trend: __R  Range: __R                  │
├─ DISCIPLINE ────────────────────────────────────────────────────┤
│ % plan-followed: ____%   Expectancy when followed: ____R          │
│                          Expectancy when broken:   ____R          │
└─ ACTION: cut ______ | fix exit/filter ______ | keep ______ ──────┘
```

E. Weekly Review

```
WEEK OF: __________   Trades: ____
Win%: ____  Total R: ____  Expectancy: ____R  Profit factor: ____
% plan-followed: ____%
Best decision: _______________________  Worst mistake: ____________________
Respected loss limits? ◻Y ◻N    Revenge/overtrade/oversize? ◻Y ◻N
Patterns noticed: ______________________________________________________
ONE adjustment for next week (one variable): __________________________
Mindset note: _________________________________________________________
```

F. Monthly System Review

```
MONTH: __________
PER-STRATEGY (one block each):
  Strategy: __________  Trades: ____  Expectancy: ____R  PF: ____  MaxDD: ____%
  Verdict: ◻ KEEP   ◻ FIX (what: ________)   ◻ KILL (why: ________)
```

```
OVERALL:
  Equity curve trend: ____________   Edge holding / decaying? ____________
  Biggest leak found (segmentation): ____________________________________
  One refinement to test next month (re-validate before trusting): ______
  Sizing: current risk ____%  → change? □ no  □ yes to ____% (justified by: ____)
  Discipline grade (process, not P/L): ____ / 10
```

G. Research / Idea Pipeline

> Keep new edge ideas here — separate from live trading — to test via the loop.

```
| Idea (one-sentence hypothesis) | Logical reason | Status (idea/backtest/forward) | Notes |
|--------------------------------|----------------|--------------------------------|-------|
|                                |                |                                |       |
```

H. The development loop (keep this in view)

```
HYPOTHESIZE → SPECIFY → BACKTEST → FORWARD-TEST → TRADE LIVE (small) →
ANALYZE (KPIs) → REFINE (one variable, re-validate) → ↺ repeat forever

Prove the edge first. Journal everything. Grade yourself on process.
Size to survive. Scale only on proof. The loop IS the edge.
```